HSLU Hochschule Luzern

PREDICTING APARTMENT RENTAL PRICES IN SWITZERLAND

DSPRO1 Project

Team: Elias Martinelli, Timo Schlumpf
Supervisor: Elena Nazarenko

HSLU Hochschule Luzern

AGENDA

1 The Problem

2 Scope Status

3 Data Source & Enrichment

4 Basline Result

5 Next steps

6 Questions & Feedback

1

THE PROBLEM

THE PROBLEM

You find a listing
3 rooms, 80 m², Zürich

CHF 2 400 / mo
Is this fair?

Why it's hard to judge

Location dependency
Same flat costs 2× more across cantons

Many interacting features
Area, floor, year built, amenities ...

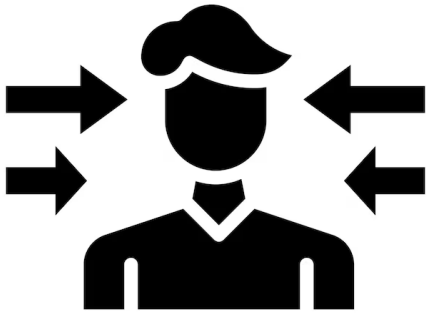
No transparent benchmark
No public reference model exists

→ We build a supervised ML model that predicts fair rent from listing data

WHY IS IT RELEVANT?

- Tenants

- Landlors



- Socially

HYPOTHESIS

Location-related variables combined with structural apartment features contain sufficient signal to predict monthly rental prices in Switzerland with practically useful accuracy and a Random Forest Regressor will outperform a Linear Regression model in doing so.

2

SCOPE STATUS

SCOPE

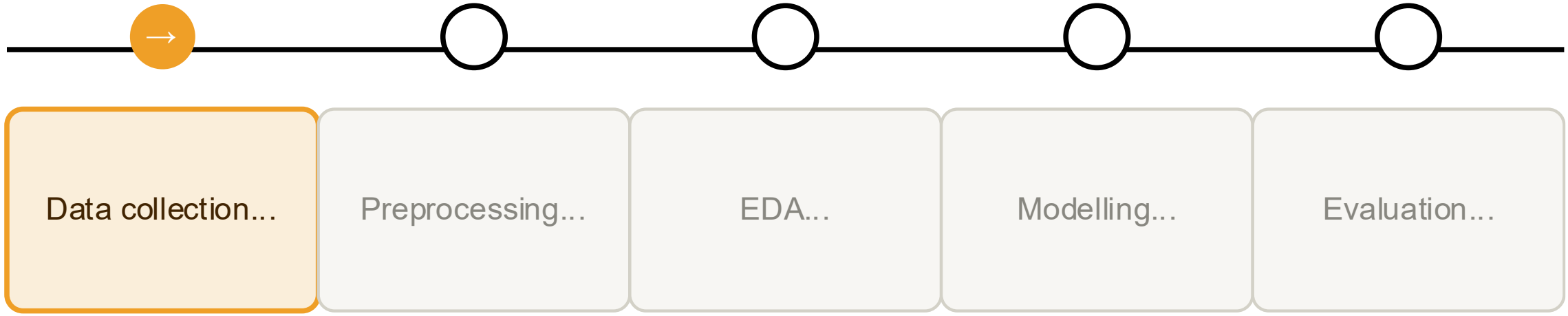
- **In Scope**
  - ✓ Web scraping of Swiss rental listings
  - ✓ Data cleaning, deduplication & outlier removal
  - ✓ Exploratory Data Analysis (EDA)
  - ✓ Feature engineering (location, structure, amenities)
  - ✓ Baseline model → Linear Regression → Random Forest
  - ✓ Model evaluation: MAE, RMSE, R^2
  - ✓ Experiment tracking with MLflow
  - ✓ Version control & Kanban via GitHub Projects

Out of Scope

- ✗ Cloud deployment or production-ready application
- ✗ Real-time or live price feed integration
- ✗ Time-series / rental price forecasting
- ✗ Non-apartment property types (houses, commercial)
- ✗ User-facing UI or web app

STATUS

Project Progression



Upcoming In progress Upcoming

PROJECT MANAGEMENT TOOL

GitHub

- Backlog
- Roadmap
- Milestones

3

DATA SOURCE & ENRICHMENT

DATA SOURCE

Webscrapping Data from comparis.ch

Wohnung mieten in der Schweiz: 27’829 Resultate

Favoriten (0)

Alle verfügbaren Wohnungen und Häuser der Schweiz auf comparis.ch. Für Sie gefunden auf comparis.ch, immobilier.ch, properstar.ch, immoscout24.ch, homegate.ch und mehr als 300 Weiteren.

Filter (3)

Suchabo anlegen

Sortieren nach:

Erstmals gefunden am

Wohnung

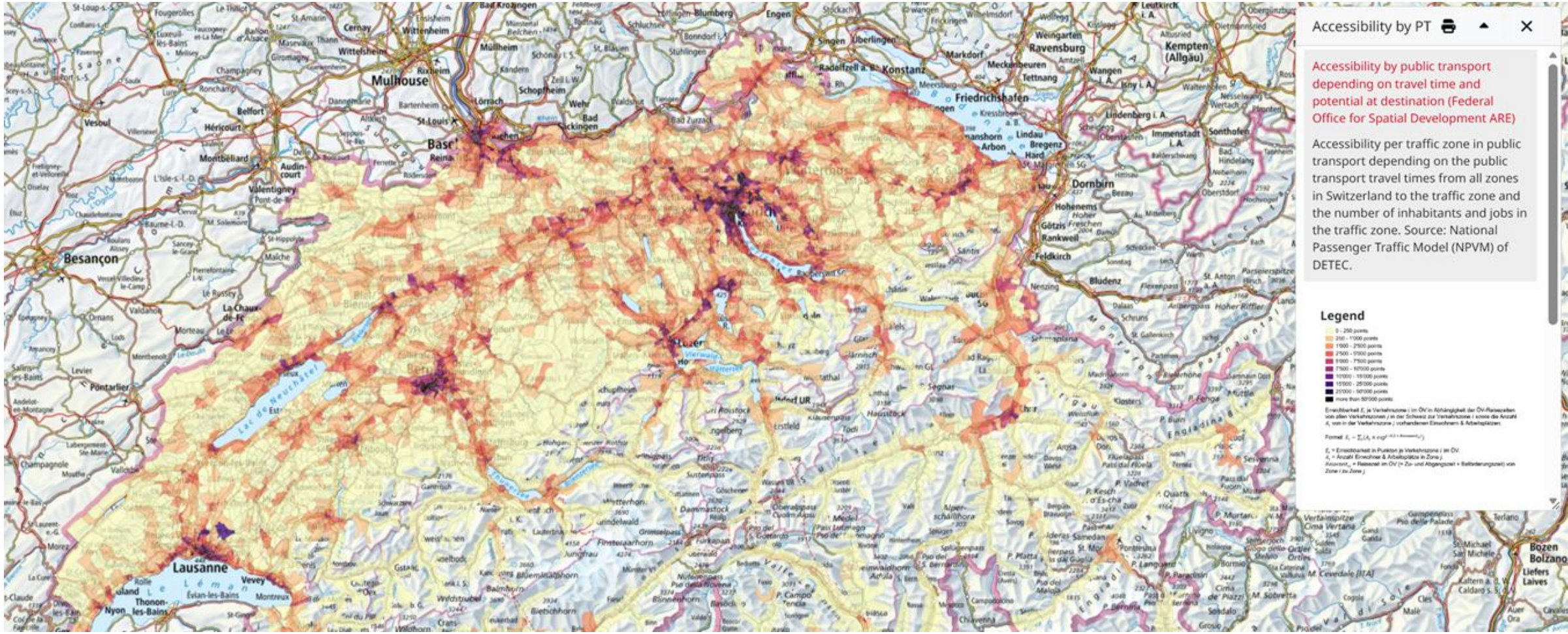
Online seit 32 Minuten

3.5 Zimmer, 76 m², 1. Etage

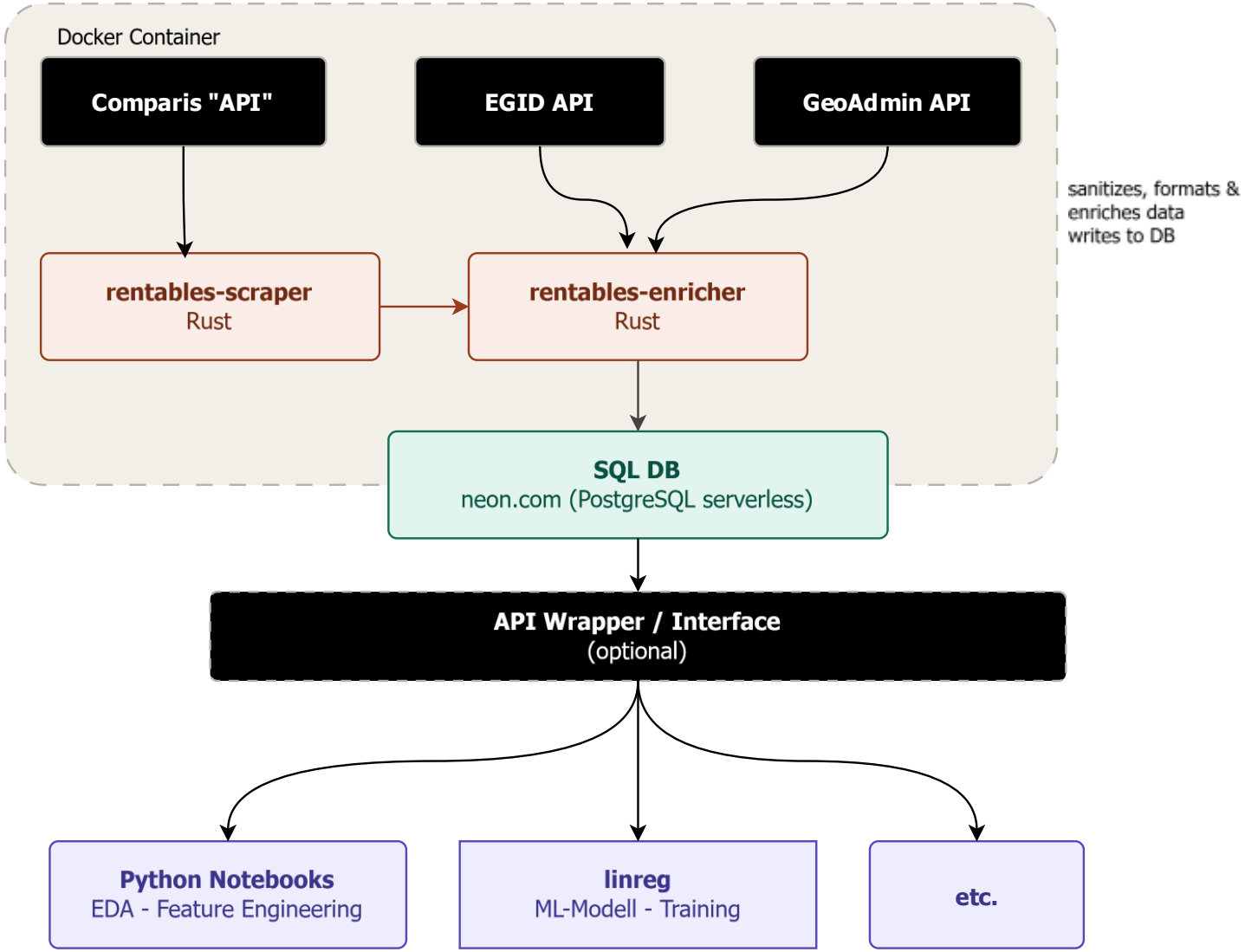
«SUPERBE APPARTEMENT MODERNE»

DATA ENRICHMENT

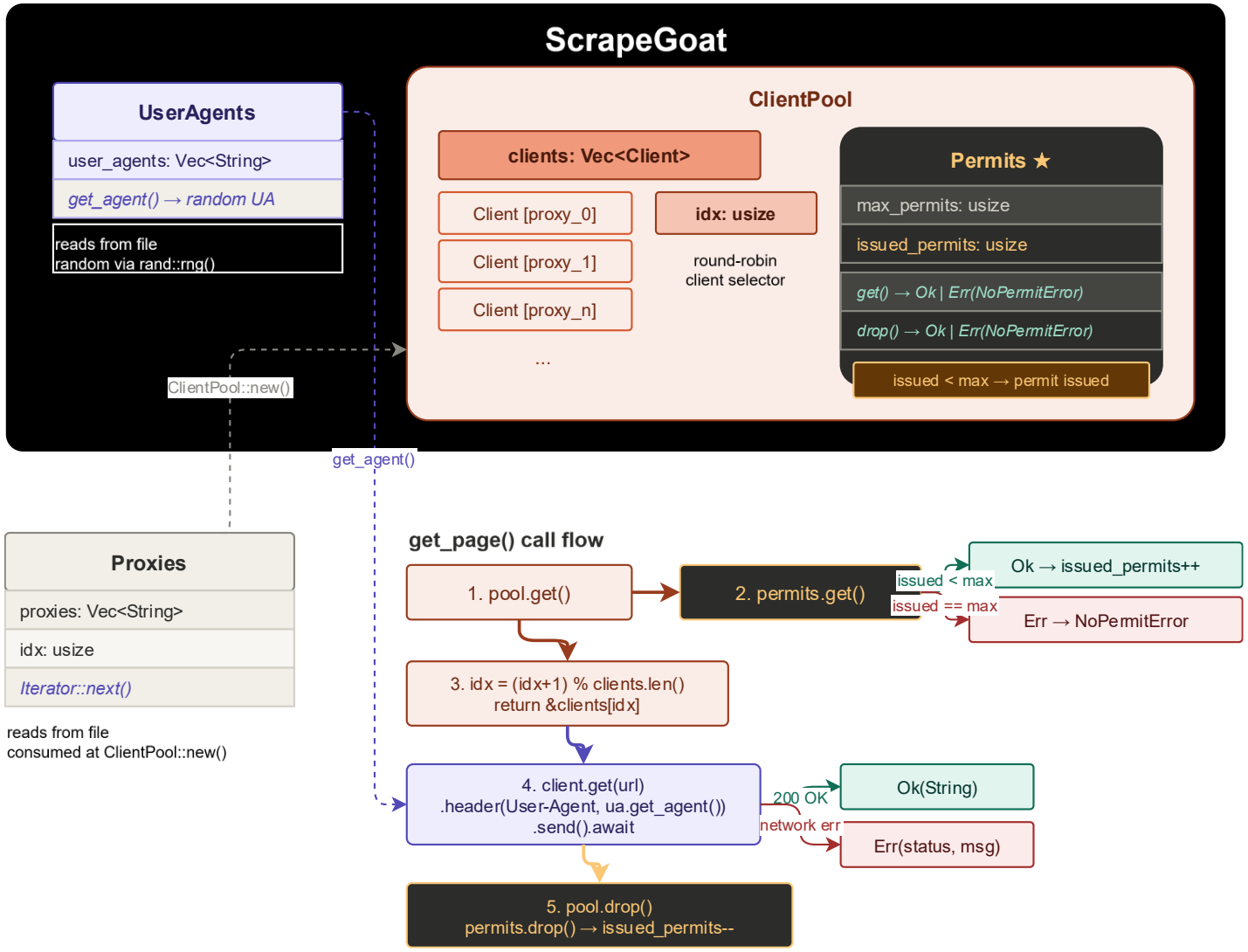
- Over API => geo.admin.ch



DATA PIPELINE HIGH-LEVEL DESIGN



POOL & PERMIT SYSTEM



4

BASLINE RESULTS

BASLINE RESULTS

Baseline strategy
Predict mean rent for every apartment regardless of features

→

Training mean
CHF 2 184
predicted for all listings

Baseline performance

MAE
CHF 612
avg. prediction error

RMSE
CHF 847
penalises large errors

R^2
0.00
no variance explained

Interpretation
Off by CHF 612 on average — this is our lower bar. Any real model must clearly beat this.

5

NEXT STEPS

FROM LISTING TO PREDICTION

Baseline
Predict mean rent
of training set
No learning needed

vs →

Linear regression
Linear feature weights
Interpretable
Fast to train

vs →

Random forest
Non-linear interactions
Tree-based ensemble
Robust to outliers

Lower bound | Interpretable model | **Our hypothesis: best**

Training set (cross-validation) | Validation | Test (hold-out)

MAE
avg. error in CHF

RMSE
penalises big errors

R^2
variance explained

MLflow tracking
params · metrics · artifacts

MODEL COMPARISON STRATEGY

Listing data
scraped from platforms

→

Preprocessing
clean · encode · scale
remove outliers

→

ML model
trained regressor
cross-validated

→

ŷ CHF

Input features

Location

- Canton
- City
- ZIP code

Structural

- Area m²
- Rooms
- Floor
- Year built

Amenities

- Balcony
- Parking
- Furnished
- Property type

Target variable: **Monthly rent in CHF (continuous)**

THANK YOU FOR YOUR ATTENTION.

We are happy to answer any questions or discuss the presented topics further.

HSLU Hochschule Luzern

DSPRO1 Project

Team:
- Elias Martinelli
- Timo Schlumpf

Supervisor:
- Elena Nazarenko

HSLU Hochschule Luzern